NutriVision – AI-Powered Indian Food Nutrition Estimator

BS IN DATA SCIENCE AND APPLICATIONS, IIT MADRAS
GROUP 2

Problem Statement

Content

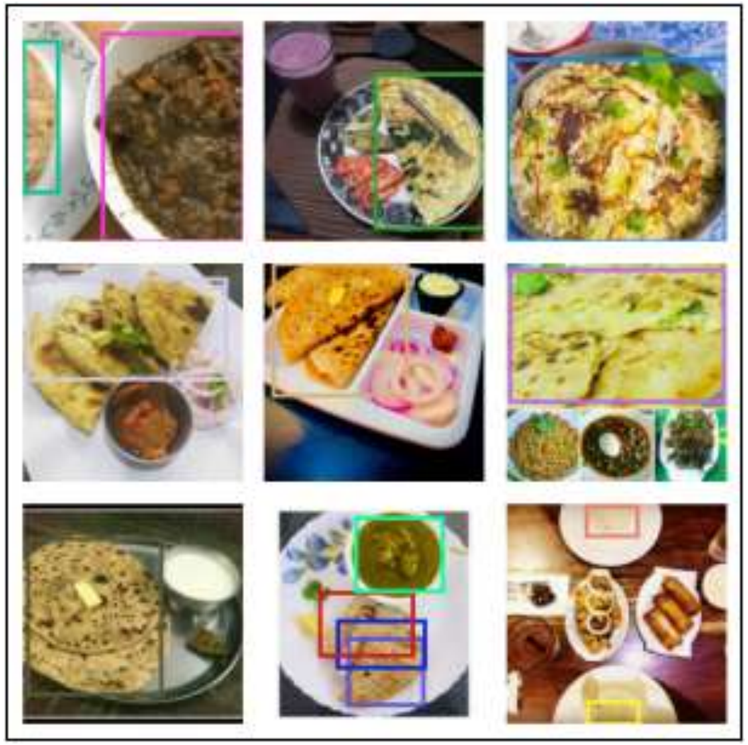
Estimating nutrition intake from meals is difficult because:

- Indian meals contain multiple food items on one plate
- Portion sizes vary significantly
- Manual calorie logging is inconvenient
- Existing apps require manual food entry

Our Solution

NutriVision uses Computer Vision + AI + RAG to automatically estimate:

- Food type
- Portion weight
- Nutritional intake
- Personalised diet suggestions



NutriVision — AI-Powered Indian Food Nutrition Estimator
Milestone 2: Dataset Analysis, EDA & System Architecture
Team: Group 2

Objective

- Build an AI system that detects Indian food from a plate image
- Estimate portion size
- Calculate nutrition values
- Provide personalized diet suggestions

The system consists of 4 major stages

1. Food Detection : Model: YOLOv8
2. Weight Estimation : Model: MLP regression model
3. Nutrition Lookup : Source: Indian Food Nutrition Dataset
4. Personalised Suggestions : RAG + Gemini + MedGemma

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📸 User photographs plate
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┌──────────────────────────────────────────┐
│ STAGE 1 — Food Detection                 │
│ YOLOv8 fine-tuned on:                    │
│   IndianFood10 + IndianFoodNet-30        │
│   + South Indian Food Detection dataset  │
│ Output: food labels + pixel segmentation masks│
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│ STAGE 2 — Weight Estimation              │
│ pixel_food ÷ plate_food = ratio          │
│ ratio + food_class+pixel_plate+pixel_food → MLP model
│                                          │
│ Output: estimated weight in grams per food │
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│ STAGE 3 — Nutrition Lookup (Rag)         │
│ food_name →                              │
│ Indian Food Nutritional Values (Kaggle)  │
│ Output: calories, protein, carbs, fat, fiber │
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│ STAGE 4 — Personalised Suggestions       │
│ Daily intake vs ICMR RDA (by age/gender) │
│ RAG: Pinecone + ICMR Guidelines PDFs     │
│ Gemini API → Indian food-aware suggestions │
│ MedGemma → medical safety review         │
│ Output: personalised dietary advice      │
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Datasets Used

Dataset	Purpose	Total Images
IndianFood10	Detect common Indian foods	**11,5**
IndianFoodNet-30	Expand food class coverage	**5,4**
South Indian Food Dataset	Balance regional cuisine diversity	**9,3**

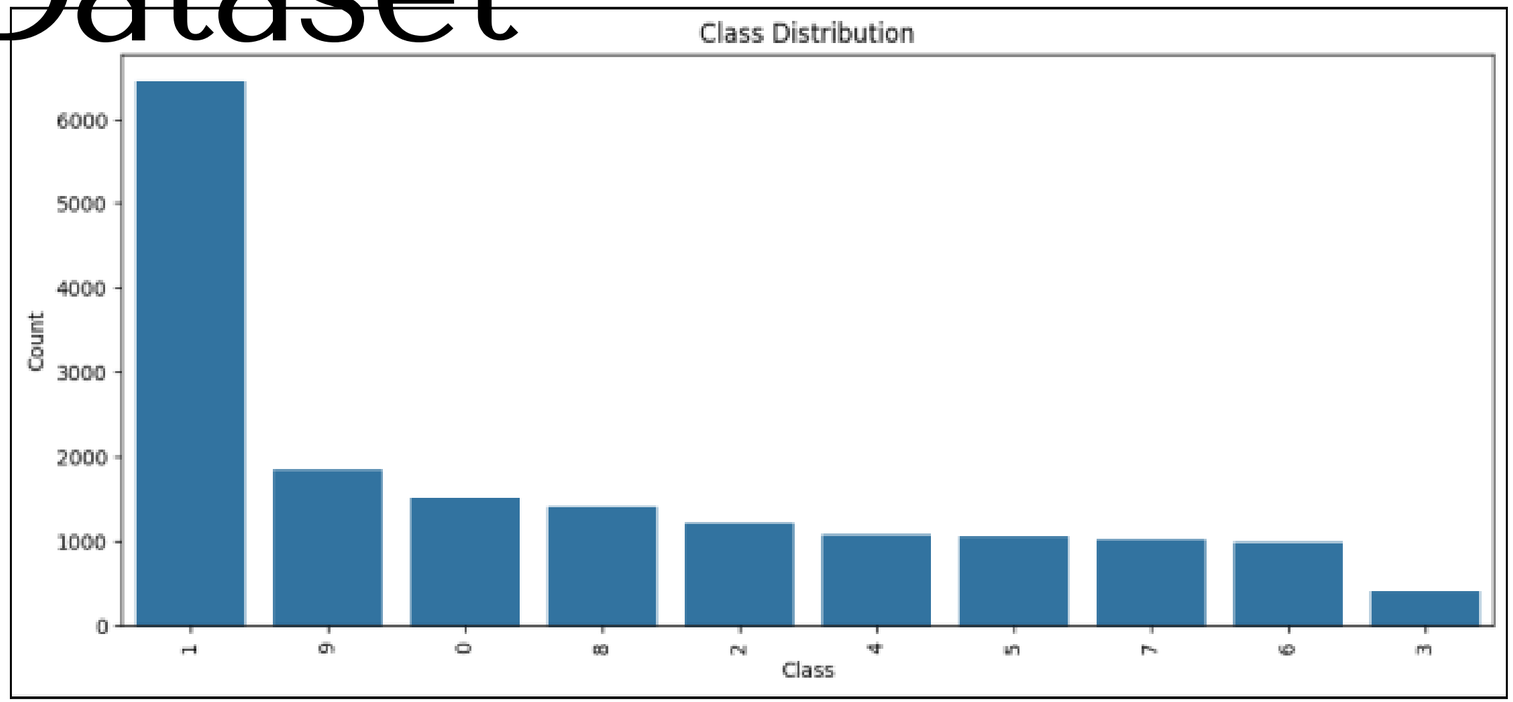
IndianFood10 Dataset

Source: Roboflow Universe

- 11,547 images
- 10 food classes
- Annotated with bounding boxes
- 842 multi-dish plates

Classes include:

- chapati
- rice
- dal
- idli
- vada
- sambar
- poha
- pongal



Problem observed:

- Class imbalance across categories: Class 1 has 6000 images
- Data augmentation for minority classes

IndianFoodNet-30 Dataset

Purpose: Increase food class coverage.

- 5,446 images
- 30 Indian dishes
- Bounding box annotations

Examples of added foods:

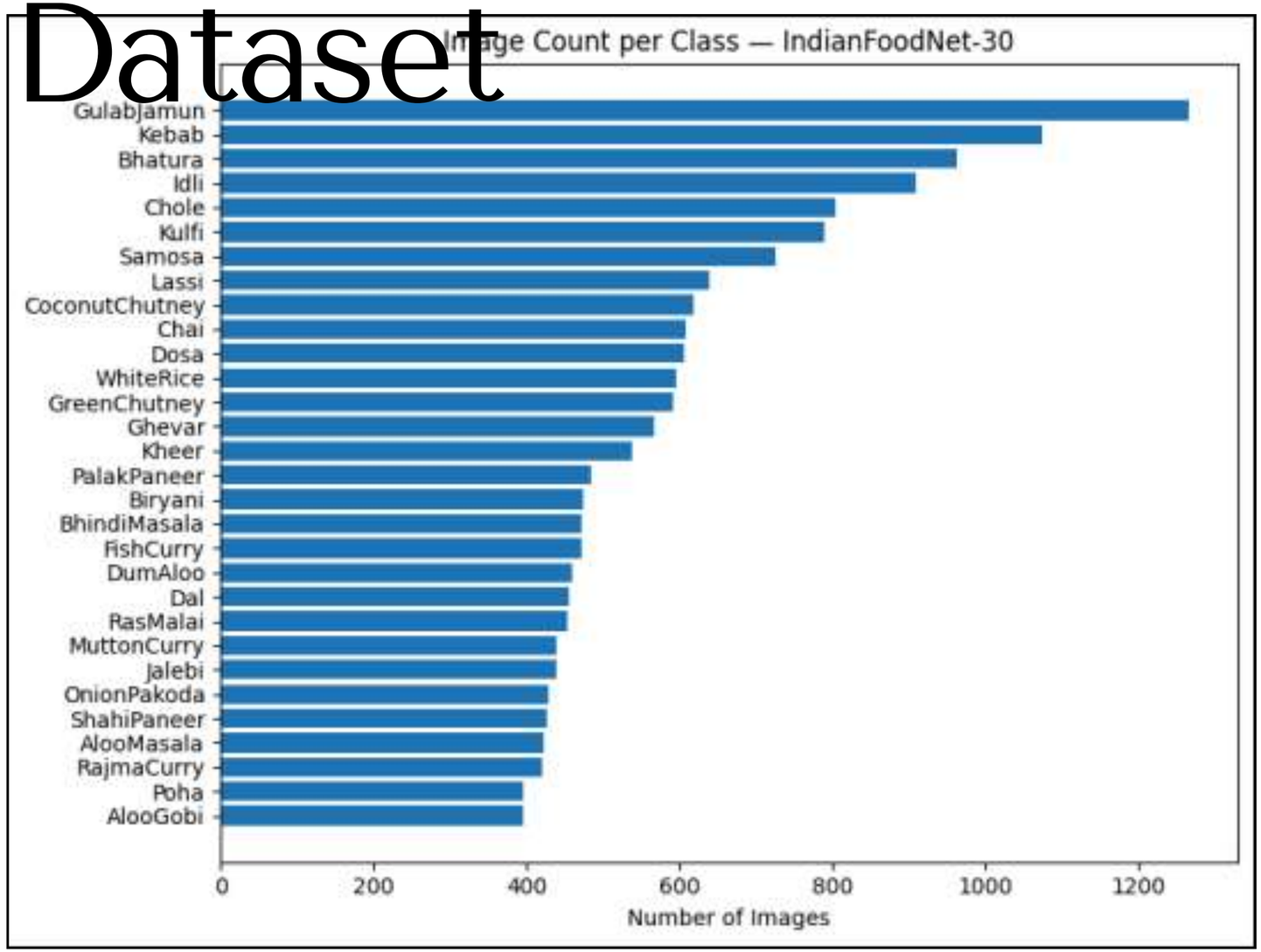
- biryani
- butter chicken
- dosa
- rajma
- aloo paratha
- chole

Problem observed:

- Class imbalance across categories

Solution:

- Data augmentation for minority classes



South Indian Food Dataset

Purpose :

- Balance regional representation.
- 9,338 images
- Focus on South Indian cuisine

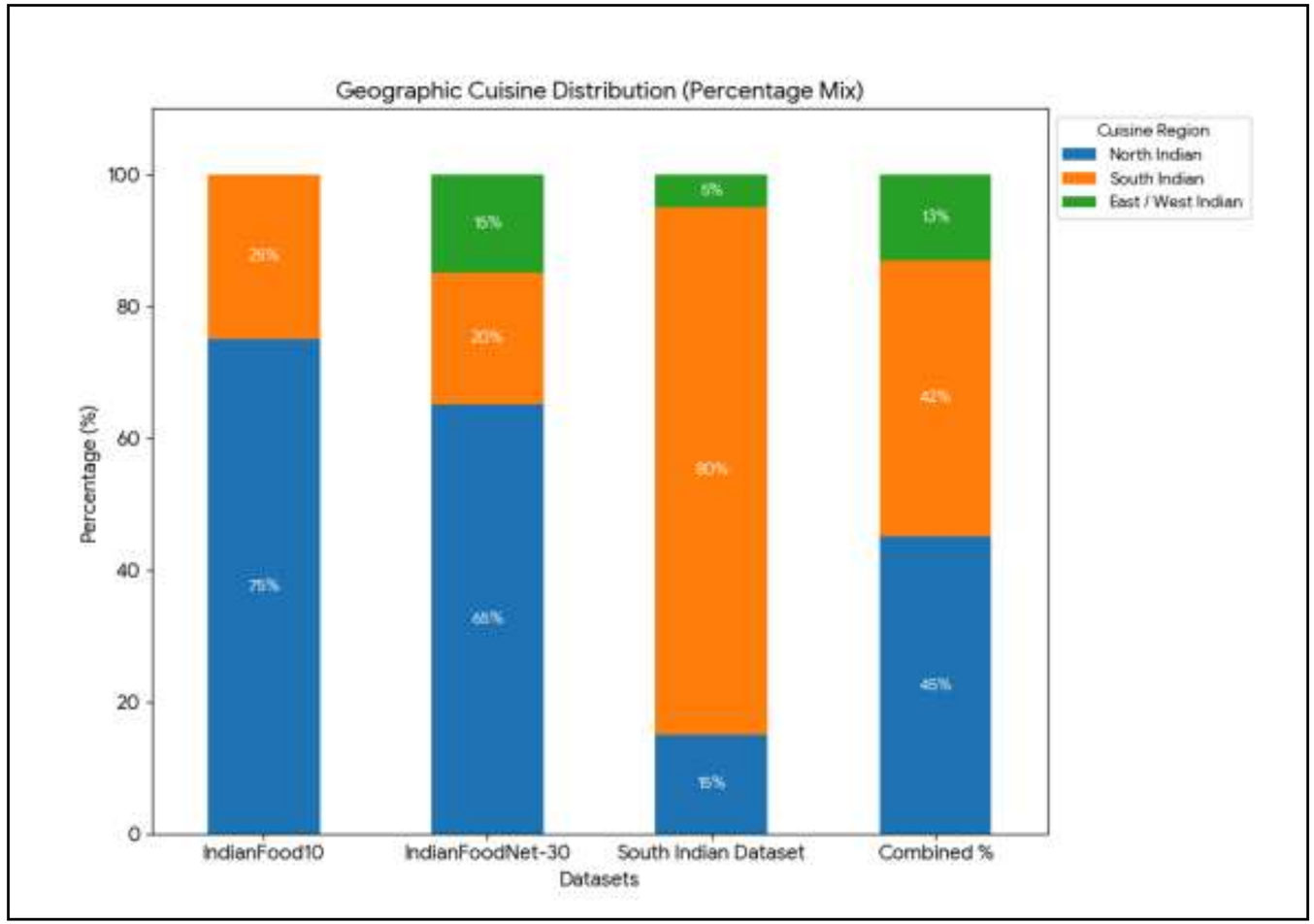
Key foods:

- dosa
- idli
- vada
- rasam
- uttapam
- payasam
- chutney

Impact

Without dataset
South Indian representation: 23%

After combining datasets
South Indian representation: 42%



Dataset Quality Analysis

Annotation quality check results:

Check	Result
Corrupted images	None found
Missing labels	None found
Annotation format	YOLO bounding box — consistent across all 3 datasets

Fig 4.1.4: Four examples of annotation quality issues — loose boxes, missing label, edge truncation

Multi-Dish Plate Analysis

Multi-dish plates are critical because:

- Real Indian meals contain multiple foods
- Model must detect multiple items simultaneously

Statistics:

- 2 foods: 68%
- 3 foods: 24%
- 4+ foods: 8%
- Most common plate combination:
  Rice + Dal + Roti

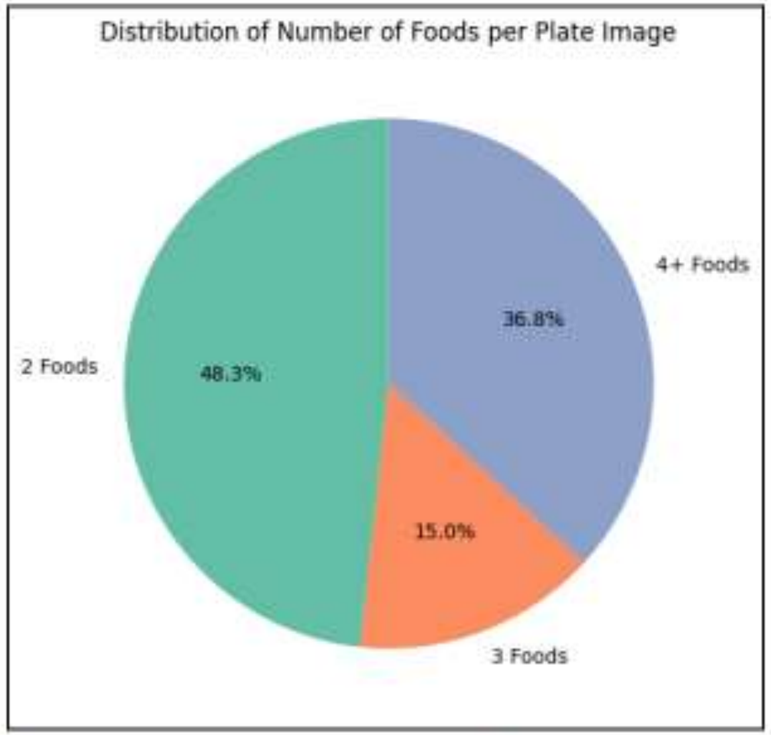


Fig 4.1.3: Distribution of number of foods per plate image (2 foods, 3 foods, 4+ foods)

Nutrition Dataset

Source: Kaggle Indian Food Nutrition Dataset

- 1014 rows
- Nutritional values per 100g

Features include:

- calories
- protein
- carbohydrates
- fat
- fiber
- sodium
- calcium
- iron

Used for nutrition lookup after weight estimation.

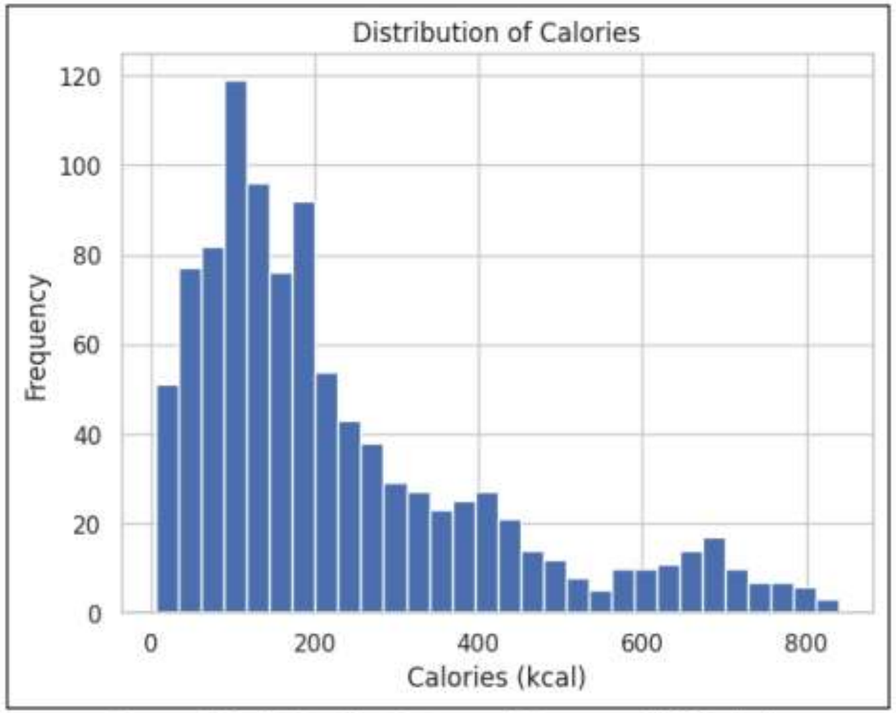


Figure 4.4.1: Calorie distribution per 100g across all 310+ foods

Weight Estimation Dataset

Problem:

No public dataset exists for food weight estimation.

Solution:

Custom dataset will be created by the team.

Dataset:

- 100 plate images
- Ground truth weight measured with kitchen scale

Pipeline:

- Capture plate image
- Run YOLO segmentation
- Calculate pixel ratio

  pixel_ratio = food_pixels / plate_pixels

- Train MLP to predict weight.

Step 1: Plate photographs collected by team (100images)
Overhead angle,kitchen scale ground truth

Step 2: Images passed through trained YOLOv8-seg model
→ food_name, pixel per food
→ plate_pixel from plate detection class

Step 3: pixel_ratio = pixel_food ÷ pixel_plate
This is the primary input feature for the MLP

Step 4: Each row saved to CSV:
[food_name,pixel_ratio,pixel_plate,pixel_food, actual_weight_g]

Step 5: MLP trained on this CSV
Input: pixel_ratio,pixel_plate,pixel_food +
food_class_encoded
Output: predicted weight in grams

RAG Nutrition Suggestion Pipeline

After nutrition calculation, the system generates diet suggestions.

Components:

- Gemini API → reasoning engine
- Pinecone → vector database
- ICMR guidelines → nutrition reference

Process:

1. Detect nutrient deficit
2. Retrieve ICMR guideline passages
3. Generate personalised recommendations

Document	Coverage	Why Included
ICMR Dietary Guidelines for Indians 2024	RDA values, food groups, and Indian meal planning	Primary reference for all suggestions

Alternative Plan

To ensure consistent model training:

- Image preprocessing:
- Resize → 640×640

Augmentation:

- brightness adjustment
- rotation
- scaling

Annotation preparation:

- segmentation masks generated using SAM-2

Preprocessing Step	Action	Reason	Applied To
Resize	640×640 px	YOLOv8 standard input resolution	All image datasets
Normalise	Pixel values 0.0–1.0	Stable training gradient flow	All image datasets
Treat missing values	Imputed missing values with the median	Data loss	Nutrition dataset

Conclusion

Key achievements of Milestone-2:

- Multiple datasets identified and analysed
- Dataset quality assessed
- Class imbalance identified and addressed
- Custom dataset created for weight estimation
- Preprocessing pipeline defined

Result:

Dataset is now ready for model training in Milestone-3.